LLM-based Personal Digital Twins: A Survey

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Abstract

Personal digital twins (PDTs) are expected to emerge as a new paradigm for representing individual intelligence through large language models and lifelong personal data. Unlike traditional digital twins or task-oriented personal assistants, PDTs aim to maintain an identity-consistent and cognition-evolving representation of an individual across time and contexts. This survey interprets why PDTs are important from views of diversity, intelligence evolution and reliable interactions, and reviews recent advances in LLM-based PDTs, focusing on three core components: personalization, memory and continual learning, and privacy and ethics. We also discuss evaluation metrics, benchmarks, application domains, and key challenges for each. Analysis of collective opinions across 10 open questions reveals that *work* scenario is where participants mostly expect twins to appear: work tasks (70.3%), remember work, study, goals (79.7%), make work, and life easier (70.3%). Privacy leaks and unauthorized actions are major causes of trust loss. Interesting, more than half participants prefer an editable twin over one that resembles themselves.

1 Introduction

Driven by sustained interest in lifelogging across formats, personal data is growing exponentially, from text and image streams on X, Instagram, WeChat Moments, and Snapchat, to long-form videos on YouTube and Bilibili, and short videos on TikTok, Kuaishou, and YouTube Shorts (De Kerckhove, 2021). Meanwhile, large language models introduce the possibility of externalizing cognitive functions such as memory, planning, and judgment (Singh et al., 2025; Comanici et al., 2025). The decision-making processes located within the human brains increasingly shift to AI systems. These data and techniques lay the groundwork for a new form of individual intelligence: internal and external intelligence re-integration within the person. We refer to it as *personal digital twin*.

Tracking from 2021 to 2026, as shown in Table 1, most surveys either focus on *(i) digital twins (DTs)* or *(ii) human digital twins (HDTs)*. The former has been extensively applied in smart manufacturing, transportation, or mobile networking simulation (Li et al., 2025d), while its paradigm is limited to embody non-living entities, e.g., robots and vehicles (Chen et al., 2023). Human digital twins are DTs adopted in human-centric systems (Chen et al., 2023). Driven by its applications in healthcare and industry, HDTs gather data to represent human bodies, dynamically reflect molecular status, physiological status, emotional and psychological status, and lifestyle evolutions (Lin et al., 2024). They are primarily physical signals of patients and workers, rather than non-physical factors like cognitive, social, cultural, and ethical dimensions of normal individuals. See more background of DT and HDT in Appendix A.

The personal digital twin (PDT) extends general HDTs to individuals, creating a digital replication based on personal data and life history (De Kerckhove, 2021). In 2021, prior to the rise of large language models, De Kerckhove (2021) conceptually discussed what PDTs are, why they are emerging, speculated the future applications, and implied their impacts from cognitive, societal and ethical views in high level. However, by 2026, how to achieve these visions in practice remains largely unresolved, though a volume of work has explored its relevant topics including LLM personalization, personal assistants, and role playing (Zhang et al., 2024; Chen et al., 2024a,b). They primarily focus on how models adapt to users. In this work, we first distinguish PDTs from personal assistants, then present three core technical components of a PDT framework with representative approaches and associated challenges for each, and finalize by 10 open questions with collective insights.

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Study	Scope	Application Domain	Survey Focus on
Wang et al. (2023b)	Internet of DTs (IoDT)	Intelligent cities	IoDT architecture with cyber-physical interactions; key characteristics and communication modes; taxonomy of security and privacy threats in IoDT
Li et al. (2025d)	Network digital twin	Mobile networks	The role of generative AI in simulating network of digital domains
Chen et al. (2023)	Human digital twins	Personalized healthcare (PH)	differences between HDT and DTs; universal framework and essential functions of HDT; requirements and challenges in PH
Lauer-Schmaltz et al. (2024)	Human digital twins	Industry, energy, smart cities, healthcare, E-commerce	definitions of DTs and HDTs and the key design considerations
Lin et al. (2024)	Human digital twin	Industry, healthcare	HDTs monitor, predict, and give suggestions to humans, based on sensing and perception data (physical signal)
Rietdijk et al. (2025)	Human digital twins	Personalized fitness coach	machine learning algorithms for HDTs in healthcare using wearable physical activity data, with a focus on personalization and model optimization
Abellán et al. (2026)	Human digital twins	Industry, healthcare	HDT concept and different aspects of a HDT considers: human dimensions, information gathered, technological frameworks, quality assessment, ethical and legal concerns.
Li et al. (2024a)	Personal LLM agent	Intelligent personal assistants	LLM agent's architecture (key components and design choices), capability, efficiency and security
Todericiu (2025)	Virtual assistant (VA)	Healthcare, education, customer service	Analysis of diverse applications of VAs, e,g, Github Copilot, Google Assistant, with emphasis of smart speakers
De Kerckhove (2021)	Personal digital twins	–	High-level conceptual framing of what PDTs are, why they are emerging, and what they imply from cognitive, societal and ethical views.
Ours (2026)	Personal digital twins	Daily usage, e.g., companion, meeting proxy	definition, key architecture components with representative methods and challenges, open questions and insights towards individual intelligence representation

Table 1: Surveys related to personal digital twins: digital twins, human digital twins, personal and virtual assistant.

2 Positions on Personal Digital Twin

A personal digital twin moves from population-level representations to the individual. It captures not only generic. shared human traits but also the subtle, fine-grained differences that distinguish each unique person (De Kerckhove, 2021). As shown in Table 1, PDTs have been relatively under-explored in the literature, while closely related research such as virtual/personal assistants and LLM-based agents has gained significant attention in recent years (e.g., OpenClaw). We identify major differences between personal digital twins, personal assistants and role-playing in our view, and interpret why the PDT is important.

2.1 Personal Digital Twin vs. Assistant

Personal assistants can be viewed as early or partial realizations of personal digital twins, yet the two differ in objectives and depth of personalization. A personal assistant is primarily task-oriented, with the main objective of optimizing user satisfaction in the moment – completing tasks, adapting behavior as needed, and responding flexibly to user preferences. Its personalization is tied to *an immediate function* aligning to the user's preference rather than a long-term identity.

In contrast, a personal digital twin aspires to be a persistent and evolving representation of an individual. It integrates not only preferences, but also values, behaviors, memories, and personality over time. Its objectives are twofold. *First,* a PDT represents and preserves a persona, with three features. *(i)* A PDT should maintain a coherent and stable identity across interactions. Meanhile it should flexibly adapt the persona to different contexts. *(ii)* A PDT should be compatible with the fast and continual growth of an individual's knowledge and experience. *(iii)* The impact of learning, experiences, and self-reflection can gradually change an individual's persona, which should be captured subtly. *Second,* similar to personal assistants, PDTs also need to perform tasks effectively. Task execution under persona-consistency constraints make it more challenging than single-objective optimization. Then, a question naturally arises: *Why is enabling a persona so important?*

2.2 Why Does Personality Matter?

We analyze the reason from three perspectives.

Diversity From the development of scaling AI paradigm, diversity of data is fundamental. PDTs offer a pathway to maintain the diversity in data.

As AI systems scale, their performance depends not just on the quantity of data, but more on its novelty and diversity. However, with the widespread use of LLMs, an increasing share of future training data will be machine-generated. These synthetic data tend to reflect average patterns: generic language, common styles, and the most probable responses, while underrepresenting the rare, novel, and unique characteristics found in pre-LLM human expression, losing the richness and variability.

Over time, this trend creates a skewed data distribution: highly diverse, creative, and unconventional human inputs become the *long tail*, receiving less weight during training. As a result, models risk becoming increasingly homogenized, losing the ability to capture nuance, originality, and rare but valuable perspectives. The problem compounds further if human behavior itself is influenced by AI, in which people may rely more on generated content, think less independently, and produce fewer distinctive ideas. In such a feedback loop, both humans and AI systems converge toward mediocrity, making it harder to drive innovation forward.

Maintaining data diversity is therefore not optional, yet essential for sustainable AI development. Ensuring that diverse, high-quality data remains central rather than marginalized in the training distribution is key to preserving creativity, fostering innovation, and enabling AI systems to continue advancing meaningfully. PDTs offer a pathway to defend aganist the generic AI systems, maintaining diversity in data and in human society.

Intelligence Evolution In the book *A Brief History of Intelligence*, five breakthroughs made our brain. LLMs primarily have captured the outputs of the second and fifth stages: (ii) reinforcement learning through trial and error (reward and penalty) and (v) the ability to use language to share knowledge and accumulate culture.

However, they lack critical components from the other stages: *(i) steering* enables animals to approach good and avoid bad, associating primitive emotions like fear with moving away; *(iii) internal simulation and planning* in brain before action (the ability to mentally simulate possible consequences and plan actions ahead); and *(iv) mentalizing* models the minds and intentions of others, advancing social intelligence and cooperation.

Because of these missing elements, current AI systems struggle to develop perceptiveness of the real world, robust reward models and effective mental simulation environments. Instead, they rely heavily on external data and real-world iterative interactions, making learning slower, more expensive, and less efficient. PDTs offer a path forward (iii) and (iv), meanwhile enabling human-like reward modeling derived from intuitions and consciousness as (i). Internal simulation within the brain could accelerate progress in reinforcement learning and allow AI systems to learn more efficiently with less dependence on costly real-world feedback.

Reliable Interaction From application view, only individuals themselves know what they want. The best alignment with a user is an AI-powered themselves, i.e., personal digital twins. PDTs optimize long-term alignment, where decisions and behaviors keep consistent with the user's values and preferences, not just immediate thoughts. That is, PDTs maintain long-term goals, rather than satisfy an immediate instruction. For example, if a user impulsively asks AI systems to draft a resignation letter late at night, a PDT may consider the user's long-term goal for financial stability and suggest drafting but not sending it until the next day. This can build more stable and reliable interactions between humans and AI systems, preparing the step stone for future brain compute interaction.

Role-playing though also involves personas, benefiting diversity, it cannot achieve intelligence evolution and reliable interactions.

2.3 Personal Digital Twins vs. Role-Playing

Role-playing adopts predefined or fictional personas (e.g., a historical figure or a movie character). These personas are typically grounded in knowledge already learned during pretraining or post-training, so models can reproduce them more reliably than PDTs given the priori. Also, when role-playing personas are creator-authored artifacts (e.g., novels, scripts, biographies, or character descriptions), the character's memories, experiences, and development are bounded by the creator's design, making them less authentic and less grounded in real individual-specific data than PDTs.

Personal digital twins model a specific individual who is often unknown to the model beforehand.

This introduces several challenges: The persona must be constructed from limited, user-specific data, rather than existing knowledge. The system must ensure consistency over time, rather than generating one-off persona-aligned responses. The persona must evolve dynamically, reflecting new experiences and updates. Role-playing techniques are insufficient to build a PDT which requires identity coherence, persistent memory, and continuous evolving, going beyond static roles.

2.4 Unique Challenges in PDTs

Compared to personal assistants and role-playing agents, PDTs introduce several unique challenges: *(i)* Persona representation and preservation: represent and learn a complex persona from limited individual-specific data; and preserve a stable personality with adaptive context-sensitive behavior; *(ii)* Two objective alignment: optimize persona consistency and task performance; *(iii)* Maintain long-term memory and update evolved user representations over time (fast cognitive versus slow personality changes); *(vi)* Privacy protection and ethical concern. In the following Sections, we will present previous representative approaches and our insights for future directions for each challenge.

3 Personalization

Personalization, the capacity of an LLM to identify, represent, and adapt to individual users, distinguishes a long-term AI partner from a simple conversational tool. We examine why personalization matters (§3.1), survey approaches for persona modeling (§3.2) and personalization techniques (§3.3), discuss evaluation (§3.4) and application scenarios (§3.5), and identify open challenges (§3.6).

3.1 Why to Personalize?

Personalization is motivated by established theories of human learning. The Zone of Proximal Development (ZPD) (Vygotsky and Cole, 1978) argues that effective instruction must match an individual's current capabilities. Cognitive Load Theory (Sweller, 2011) similarly suggests that the same information may be either accessible or overwhelming depending on prior knowledge.

Empirical studies further show that personalization improves user engagement and trust. Yoon and Lee (2021) found that perceived empathy in AI systems increases user acceptance. Users form relationship-like bonds with chatbots that can remember prior interactions (Pentina et al., 2023). Zhang et al. (2025d) further found that appropriately calibrated long-term interactions can positively affect well-being. However, current LLMs do not personalize reliably by default.

LLM outputs are often implicitly calibrated towards dominant user groups, typically educated English-speaking users (Santurkar et al., 2023). As a result, performance can degrade for vulnerable populations, older adults, or users with different linguistic and cultural backgrounds (Poole-Dayan et al., 2024; Valtolina et al., 2024; Albdrani and Al-Shargabi, 2023). Recent studies show that models alter response quality and factual behavior based on perceived user identity and language style (Kearney et al., 2025; Lee et al., 2026). Similarly, in Bourdieu's terms (Bourdieu, 1991), linguistic capital increasingly influences access not only to institutions, but also to AI-generated guidance. Therefore, we expect to calibrate AI systems to represent diverse personas via personal digital twins.

We distinguish three interconnected layers of personalization, illustrated in Table 2:

❍ **Persona modeling** constructs and updates representations of the user. Its primary goal is to *understand and represent* the user (§3.2).

❍ **Personalization logic** is the adaptation layer that leverages these representations to modify system behavior for a specific user, adjusting tone, content selection, difficulty progression, or recommendation strategy. It addresses: *how should the system behave differently for this user?* (§3.3).

❍ **Personalized application** is the complete product or system built around these capabilities, integrating persona modeling, personalization logic, memory, and user-facing interfaces (§3.5).

Distinguishing these layers clarifies both technical contributions and failure modes: a system may model the user accurately yet personalize poorly (e.g., retrieving the right profile but failing to adapt response appropriately), or personalize effectively in isolation yet fail as an application (e.g., lacking the memory infrastructure to maintain consistency across sessions).

3.2 Persona Modeling

Unlike personalization, which adapts system behavior, persona modeling concerns *what the system knows about the user*. Existing approaches evolve from static user-provided profiles to dynamic representations inferred from interaction history.

Layer	Example: AI Learning Platform
Persona Modeling	Infers that the user is a visual learner, prefers short sessions, experiences math anxiety, and responds well to encouragement.
Personalization	Shows diagrams first, limits lessons to 10 minutes, adjusts tone to be more supportive, and adapts difficulty progression.
Personalized Application	The full platform that tracks long-term learning progress, adapts curriculum sequencing, customizes the dashboard, schedules review reminders, and evolves interaction style over months of use.

Table 2: Three layers of personalization illustrated with an AI learning platform. Persona modeling represents the user; personalization adapts behavior; the personalized application is the full adaptive product.

Static persona profiles. Early approaches rely on explicit textual descriptions of the user. Zhang et al. (2018) introduced PersonaChat, where dialogue agents are conditioned on predefined persona sentences. While effective for controlled dialogue generation, static profiles require manual specification, cannot evolve over time, and capture only limited aspects of user behavior.

Implicit persona detection. Subsequent work infers persona attributes directly from dialogue history rather than requiring predefined profiles. Cho et al. (2022) proposed a personalized dialogue system that extracts implicit persona signals from prior interactions, enabling persona construction from naturally occurring user behavior.

Dynamic and life-long persona updating. More recent work models personas as evolving states rather than fixed profiles. Wang et al. (2024d) introduced AI Persona, which continuously updates user representations from new interactions over time. This addresses a key limitation of earlier approaches: real users change as their interests, expertise, and preferences evolve. Chen et al. (2024a) characterize this transition from persona-as-input to persona-as-learned-state as a central trend in personalization research.

User embeddings. An alternative to symbolic persona representations is to model users as dense embeddings learned from interaction history. Ning et al. (2025) proposed User-LLM, which learns compact user representations for efficient personalization. Embedding-based methods are computationally efficient because their size does not grow with interaction history, but they sacrifice interpretability: it is difficult to inspect or edit what the model has inferred about a user.

Relationship to personalization. Persona modeling and personalization play complementary roles: persona modeling represents the user, while personalization adapts system behavior based on that representation. Accurate user models alone do not guarantee effective personalization, and sophisticated adaptation strategies are ineffective without reliable user representations. The methods in this subsection provide the representational foundation; the next subsection (§3.3) discusses how systems act upon these representations.

3.3 Personalization Methods

Given a user representation constructed through persona modeling (§3.2), personalization methods determine how the system adapts its behavior to a specific user. We organize approaches from inference-time adaptation to training-time personalization, following recent surveys (Zhang et al., 2024; Chen et al., 2024b; Li et al., 2025e). Table 3 summarizes the main trade-offs.

In-context learning (ICL). ICL methods inject user-specific information into prompts at inference time without modifying model parameters. Salemi et al. (2024) showed that retrieving user-authored content as in-context examples enables personalized generation, while Bao et al. (2024, 2025) extended this idea to real-time recommendation with evolving user preferences. He et al. (2024) introduced context steering for controllable personalization at inference time. ICL methods support zero-shot adaptation without per-user training, but remain constrained by context window limits and KV-cache scaling costs.

Retrieval-augmented personalization. When user history exceeds the context window, retrieval-augmented generation (RAG) retrieves relevant user information on demand. Shi et al. (2025a) extend this paradigm by incorporating collaborative filtering signals into retrieval. Li et al. (2025e) survey approaches spanning RAG-based to agent-based personalization. The main limitation of retrieval-based methods is their dependence on retrieval quality: irrelevant or missing history directly degrades downstream personalization.

Parameter-efficient fine-tuning. Unlike ICL and RAG, parameter-efficient fine-tuning internalizes user-specific patterns into model parameters. Hu et al. (2022) introduced LoRA for lightweight per-user adaptation, while Wang et al. (2024a) proposed LoRA-Flow for dynamically combining multiple user-specific modules during generation. Jang et al. (2023) introduced Personalized Soups for post-hoc merging of personalized parameter sets. These approaches enable deeper behavioral adaptation than prompt-based methods, but face scalability and deployment challenges when serving large numbers of users.

Personalized alignment. Standard RLHF aligns models to aggregate preferences, but individuals often diverge from population-level behavior. Li et al. (2025f) showed that individual-level preference data improves per-user alignment on subjective tasks. DPO (Rafailov et al., 2023) further simplifies preference optimization and facilitates incorporating user-specific feedback. Shi et al. (2024) extend this direction by learning from real interaction signals such as edits, regeneration requests, and conversation abandonment. A persistent challenge for alignment-based personalization is the scarcity of reliable per-user preference data.

Multimodal and agent-based extensions. Personalization increasingly extends beyond text to multimodal behavioral signals. Wu et al. (2024b) survey multimodal personalized LLMs incorporating visual, behavioral, and physiological information. Agent-based systems further integrate persona modeling, memory, tool use, and personalization within unified architectures. Westhäußer et al. (2025) show that persistent memory and dynamic user profiles support long-term personalized interaction across sessions, while Xu et al. (2026b) survey the emerging landscape of personalized LLM agents.

3.4 Personalization Evaluation

Evaluating personalization is fundamentally different from evaluating generic language model capabilities, due to three structural properties. First, the *absence of universal ground truth*: a response ideal for one user may be inappropriate for another, echoing the insight from complementary-label learning (Ishida et al., 2017) that supervision can come from what is *wrong* for a user rather than a single correct answer. Second, the *cold-start problem*: new users with no interaction history cannot be evaluated against personal baselines. Third, the *temporal dimension*: personalization quality must be assessed over interaction trajectories, not at single time points, because a system that adapts well initially but drifts over weeks is not truly personalized. These challenges have motivated the development of specialized benchmarks. Table 4 summarizes the major benchmarks for personalization evaluation.

Metrics for personalization quality. Traditional NLG metrics such as BLEU and ROUGE measure textual overlap but are largely insensitive to user-specific appropriateness. Dudy et al. (2021) argue that evaluation should shift from generic relevance to user-relevance, while Nimah et al. (2023) show that standard metrics often disagree with human judgments when output quality depends on user-specific factors. Recent benchmarks increasingly evaluate dimensions beyond generic response quality, including user-specificity, consistency across interactions, adaptiveness to evolving preferences, and fairness across user groups. Zhao et al. (2025a) show that personalization systems can systematically favor users with richer interaction histories, while Tan (2025) demonstrate that underrepresented behavioral patterns can lead to a persistent "personalization gap."

Human and implicit feedback. Predefined metrics cannot capture all aspects of personalization quality. Shi et al. (2024) show that signals collected during real interactions, including edits, regeneration requests, and conversation abandonment, can serve as both personalization and evaluation signals. The LLM-as-Judge paradigm scales more efficiently than human evaluation, but its reliability for personalization remains unclear because judges may prioritize generic response quality over user-specific appropriateness (Chiang et al., 2024). To address this limitation, Chang et al. (2025) propose ChatBench, a human–AI collaborative evaluation framework that evaluates personalization across multi-session interactions.

3.5 Personalized Applications

Personalized applications represent end-to-end systems that integrate user modeling, personalization, memory, and interaction interfaces. We organize applications by stakes and personalization depth, since techniques that are appropriate in one domain may be unsafe in another.

Healthcare. Healthcare requires deep personalization under strict accuracy and safety constraints. Lammert et al. (2024) demonstrate expert-

	ICL / Prompt	RAG-based	Fine-tuning (LoRA)	RLHF / DPO
Adaptation Depth	Surface (style, context)	Moderate (knowledge)	Deep (behavior)	Deep (values)
Training Required	No	No (retriever optional)	Per-user adapter	Preference data, Train
Scalability	High	High (external memory)	Challenging	Challenging
Strength	Zero-shot, real-time	Large user histories	Complex style adaptation	Value alignment
Limitation	Context window bound	Retrieval quality	Deployment overhead	Data scarcity

Table 3: Comparison of core personalization methods. All methods assume a user representation (from persona modeling) as input and differ in how they translate that representation into adapted system behavior.

Benchmark	**Data Source**	**Domain / Scenario**	**Scale**	**Metrics**	**Major Challenge Exposed**
LaMP (Salemi et al., 2024)	User-generated content (reviews, emails, articles)	Personalized text generation and classification	8 tasks, thousands of user profiles	Accuracy, F1, ROUGE, MAE	Models struggle to leverage user history for generation tasks more than classification tasks
LongLaMP (Kumar et al., 2024)	Extended user histories from LaMP	Long-form personalized generation	4 tasks, longer output	ROUGE, personalization gain over non-personalized baseline	Performance gains diminish as output length increases; long-form personalization remains difficult
PersoBench (Afzoon et al., 2024)	Synthetic multi-dimensional user profiles	Multi-dimensional response evaluation	Multiple persona dimensions	Persona consistency, response quality, multi-dimension coverage	Single-score evaluation masks trade-offs across persona dimensions
PersonaLens (Zhao et al., 2025b)	Task-oriented assistant interactions	Conversational AI assistants	Multi-session dialogues	User-specificity, consistency, task success, adaptiveness	Models often prioritize task completion over persona-consistent behavior
Know Me (Jiang et al., 2025a)	Large-scale user interaction logs	Dynamic user profiling	Thousands of users at scale	Profiling accuracy, response personalization, temporal adaptation	Static user profiles degrade rapidly; dynamic profiling is necessary but under-explored
Personalized Deep Research (Liang et al., 2025)	Multi-step research tasks	Personalized deep research	Multi-step task sequences	Step-level and end-to-end personalization, research quality	Personalization in multi-step reasoning amplifies early-stage errors

Table 4: Representative benchmarks for personalization evaluation.

guided LLMs for precision oncology, while Holderried et al. (2024) show that personalized simulated patients improve medical training outcomes. Farhadi and Mozayeni (2025) identify challenges in grounding personalized recommendations in clinical guidelines, particularly for rare or complex patient profiles. Aththanagoda et al. (2025) further show that patient-conditioned LLMs outperform generic medical QA systems, especially for atypical cases.

Mental health. Mental health applications require sustained emotional sensitivity and longitudinal user modeling. Stade et al. (2024) propose a staged roadmap for deploying LLMs in behavioral healthcare, while Hua et al. (2025) review applications ranging from CBT support to crisis detection. Boit and Patil (2025) show that persistent user-state tracking improves continuity across sessions. However, personalization also raises risks of emotional dependency and social withdrawal in companion-style systems (Bhattacharjee et al., 2025; Reddy P et al., 2024; Malfacini, 2025).

Education. Educational applications must balance personalization with pedagogical alignment. Park et al. (2024b) demonstrate tutoring systems that adapt explanations based on inferred learner state, while Reddig et al. (2025) show that LLMs can generate personalized tutoring feedback comparable to expert-written responses. However, Sonkar et al. (2024) identify a “student data paradox,” where excessive adaptation to individual error patterns can reinforce misconceptions rather than improve learning outcomes. Razafinirina et al. (2024) further highlight persistent challenges in adaptive scaffolding and reliable student-state modeling.

Finance and law. Financial and legal applications operate under strict regulatory constraints and require grounding in authoritative knowledge. Takayanagi et al. (2025) introduce FinPersona for personalized financial advising, while Dai et al. (2026) highlight the difficulty of inferring implicit user context in high-stakes financial reasoning. Cagle and Ahmed (2024) argue that enterprise applications require domain-specific reasoning patterns beyond generic helpfulness. In law, Dehghani et al. (2025) survey LLM applications, and Cui et al. (2023) show that effective legal personalization depends on grounding in external legal knowledge bases.

Writing, coding, and creative assistance. These lower-stakes domains emphasize stylistic and workflow personalization. Nazim et al. (2025) show that EFL students benefit from writing assistance adapted to proficiency level and feedback granularity. Eshraghian et al. (2025) further show that developer satisfaction with GitHub Copilot depends strongly on adaptation to individual coding patterns and conventions.

E-commerce and customer service. E-commerce applications focus on real-time preference adaptation and relational interaction. Dam et al. (2024) survey the shift from rule-based chatbots to generative personalized assistants, while Bao et al. (2024, 2025) demonstrate real-time recommendation based on evolving interaction history. Youn and Jin (2021) and Vebrianti et al. (2025) show that relational personalization improves customer loyalty and engagement. However, Kim et al. (2025) find that undisclosed AI identity can erode trust even when personalization improves user experience.

Cross-cultural personalization. Cross-cultural personalization requires adapting not only language but also culturally embedded norms such as politeness, humor, and directness. Singh et al. (2024) study intralingual cultural adaptation, while Rao et al. (2025) show that current models generalize unevenly across cultural contexts. Amin (2024) warn that personalization systems may default toward dominant cultural norms, potentially reducing cultural diversity in professional and educational settings.

3.6 Challenges and Future Directions

We identify five major challenges facing personalized LLM systems, together with emerging mitigation directions.

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Challenge 1: Personalization amplifies hallucination and epistemic enclosure. Personalized systems may generate hallucinations aligned with a user's existing beliefs, making false content more persuasive and harder to detect. Sun et al. (2026) show that personalized hallucinations vary systematically across user profiles, while Jacob et al. (2025) describe this as a "chat-chamber effect" in which systems reinforce existing viewpoints. Related work further connects personalization to filter bubbles and reduced epistemic diversity (Lazovich, 2023; Sharma et al., 2024b; Kelley and Riedl, 2026). Promising mitigation approaches include separating factual grounding from personalized delivery, presenting alternative viewpoints proactively, and evaluating factual robustness across diverse user profiles.

Challenge 2: Sociolinguistic bias transforms into personalized prescription. Personalization systems that adapt to user language patterns risk amplifying sociolinguistic bias. Hofmann et al. (2024) show that users writing in African American English receive systematically different responses in domains such as employment and legal advice, while Bassignana et al. (2026) find similar disparities across socioeconomic registers. These findings suggest that personalization can transform descriptive linguistic signals into prescriptive quality differences. Mitigation requires separating personalization from sociolinguistic stereotyping through fairness-aware adaptation and evaluation across demographic and linguistic groups (Zhao et al., 2025a; Sorensen et al., 2024).

Challenge 3: Persona-based vulnerabilities expand the attack surface. The more faithfully a system adapts to user personas, the more attack vectors it exposes. Shah et al. (2023) show that adversarial persona descriptions can bypass safety filters, while Tang et al. (2025) identify character hallucination as an additional jailbreak vector.

Deshpande et al. (2023) further show that persona conditioning itself can increase toxic generation. Together, these findings suggest that deep personalization introduces latent safety risks beyond standard prompting attacks. Safety constraints should remain invariant across persona states, with architectural separation between personalization modules and safety enforcement mechanisms.

Challenge 4: Long-term coherence degrades in extended conversations. Personalized systems must operate across long, non-linear interaction histories containing topic shifts, revisited contexts, and conflicting instructions. Oni (2025) identify a memory–coherence tension in context-aware chatbots, where accumulated history increases interference from stale or irrelevant information. Simhi et al. (2026) further show that conversational history can trap models in suboptimal generation regions, while Gupta et al. (2024) and Laban et al. (2025) document systematic degradation in multi-turn settings. Hierarchical memory systems, intent-aware retrieval, and multi-session evaluation protocols may help reduce long-term interference and coherence drift.

Challenge 5: The right to forget conflicts with deep personalization. Effective personalization depends on integrating user data into model behavior, but privacy regulations such as GDPR require that users be able to delete their data. Zhang et al. (2025a) identify a fundamental tension: once user information has influenced model parameters, retrieval indices, or memory structures, selective removal becomes difficult. Recent work on machine unlearning attempts to address this problem (Wang et al., 2024f; Parii et al., 2025; Xu et al., 2026a), but current approaches remain incomplete and can degrade retained capabilities. To mitigate, external and modular memory architectures may make deletion more tractable than deeply internalized personalization, though reliable unlearning remains an open problem.

Future directions. Four research frontiers emerge. First, personalization-aware evaluation must jointly assess utility, factuality, fairness, and safety under personalization. Second, multi-turn robustness should be treated as a prerequisite for sustained personalization rather than a separate robustness problem (Gupta et al., 2024; Laban et al., 2025). Third, longitudinal studies are needed to understand the long-term effects of personalized LLMs on well-being, autonomy, and information diversity (Zhang et al., 2025d; Malfacini, 2025). Finally, pluralistic personalization requires systems that maintain equitable quality and epistemic diversity across user populations (Sorensen et al., 2024; Zhao et al., 2025a).

4 Memory and Continual Learning

Memory and continual learning are what separate a persistent personal digital twin (PDT) from in-the-moment personalization: they let a PDT preserve and gradually evolve a coherent persona, and improve its task behavior from accumulated personal experience without drifting from that persona. Building on recent work on agent memory and continual learning for LLMs (Hu et al., 2025; Shi et al., 2026), we distinguish PDT memory from existing LLM and agent memory (§4.1), then discuss what a PDT must remember (§4.2), how it is accessed ((§4.3), continually learned and updated (§4.4), and evaluated (§4.5), and key open problems (§4.6).

4.1 From LLM Memory to PDT Memory

Memory in a PDT builds on, but is not exhausted by, the trajectory of memory research for LLMs and agents. For LLM memory, long-context modeling and retrieval-augmented generation provide two intertwined ways of making past information available to the model (Li et al., 2024b; Liu et al., 2025; Gao et al., 2023b). The former keeps more history within the context window for direct attention, while the latter externalizes history into stores that can be queried and reintroduced on demand (Dong et al., 2024; Laban et al., 2026; Liu et al., 2024; Lewis et al., 2020; Edge et al., 2024; Gutiérrez et al., 2024). Recent work on agent memory marks a further shift from making memory available to actively managing it, deciding what should be written, retrieved, updated, or forgotten across sessions (Packer et al., 2023; Xu et al., 2025b; Chhikara et al., 2025; Lin et al., 2026). Yet even with this shift, memory is still primarily used to support the current interaction, including response generation, conversational continuity, and task performance (Hu et al., 2025; Zhang et al., 2024; Wang et al., 2024b).

This remains insufficient for the two objectives of PDTs introduced in §2.1: ($\mathcal{O}_1$) the PDT must establish, preserve, flexibly express, and gradually evolve the persona of a particular individual; and

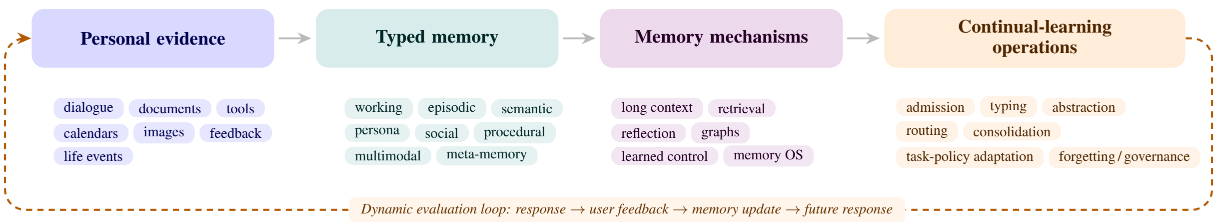

Figure 1: From personal evidence to evolving PDT behavior. Existing systems often optimize access, retrieval, or personalization separately; PDTs organize these stages into a *governed closed loop* in which memory updates affect future identity expression and task behavior.

Layer	**Representative anchors**	**Gap for PDTs**
Base LLMs	Lost-in-the-Middle (Liu et al., 2024); MemGPT (Packer et al., 2023); LongMemEval (Wu et al., 2025a)	Longer access, but no identity semantics.
LLM agents	Personal LLM Agents (Li et al., 2024a); MemoryBank (Zhong et al., 2024); MemOS (Li et al., 2025h)	Task continuity, but not durable identity grounding.
Persona-aware systems	PersonaChat (Zhang et al., 2018); LaMP (Salemi et al., 2024); PersonaLens (Zhao et al., 2025b)	User adaptation, but often static or task-local.
PDT-oriented work	BehaviorChain (Li et al., 2025c); TwinVoice (Du et al., 2025); Persistent Personas (de Araujo et al., 2026)	Identity fidelity, evolution, task utility, and governance must be jointly modeled.

Table 5: Representative memory-related work across system layers. PDTs require more than long-context access or task memory: memory must become an evidence-grounded, evolving, and governable identity state.

($\mathcal{O}_2$) it must complete tasks in ways that satisfy the user, remain consistent with that persona, and continually improve from accumulated personal experience. These objectives require a different view of memory: a PDT should not only condition responses on memory, but also let that memory evolve from accumulated experience in a way that keeps the persona coherent over time. In this view, memory shifts from a passive archive of past interactions to an identity-bearing substrate for continual learning: a PDT uses its persona and memories to respond, and each interaction provides new evidence that updates memory and gradually adjusts its representation of the user. This shift raises the central question for PDT memory: ***how should a PDT continually learn and update its memory while keeping the persona coherent?***

4.2 What a PDT Remembers

Recent work organizes agent memory from a functional perspective into working memory, experiential memory, and factual memory (Hu et al., 2025). Working memory actively curates the task context, including current goals, relevant interaction history, retrieved user facts, temporary constraints, and task-specific instructions (Yu et al., 2026; Zhang et al., 2025e; Li et al., 2025b). In PDTs, such transient context should guide the current interaction without being consolidated into stable persona preferences. Unlike conventional experiential memory, which stores reusable lessons from past interactions as a non-parametric substrate for continual improvement (Cao et al., 2025; Zhao et al., 2024; Shinn et al., 2023), experiential memory in a PDT must support improvement while remaining persona-consistent, thereby supporting $\mathcal{O}_2$. Recent work on factual memory is gradually moving toward the $\mathcal{O}_1$ goal of PDT memory by storing relatively stable or slowly evolving user knowledge, including background facts, preferences, values, relationships, routines, and salient life events (Kang et al., 2025; Wang et al., 2025a; Rasmussen et al., 2025; Tan et al., 2025).

PDT memory often requires these three types to operate jointly rather than as isolated stores. For example, when a user makes an angry, high-stakes decision after a frustrating event, a PDT should use working memory to recognize the transient emotional context, factual memory to recall the user's long-term values and persona, and experiential memory to respond in a way that is consistent with the user's stable self rather than the immediate impulse, so that the three jointly serve $\mathcal{O}_1$ and $\mathcal{O}_2$.

Memory element	Stored where	Representative mechanisms	PDT-specific role
Working memory	Context window, scratchpad, KV/activation state.	Long-context prompting; MemGPT-style context management (Packer et al., 2023); short-term memory in MemoryOS (Kang et al., 2025).	Tracks the current task, active intent, and temporary constraints; should not directly become persona evidence.
Episodic memory	Dialogue logs, vector databases, timestamped event stores.	MemoryBank-style store–recall–update (Zhong et al., 2024); LoCoMo (Maharana et al., 2024); LongMemEval (Wu et al., 2025a).	Records personal experiences, interactions, decisions, and outcomes; supports temporal reasoning over a user's trajectory.
Semantic persona memory	User profiles, persona summaries, user embeddings, adapters.	LongLaMP (Kumar et al., 2024); personalized preference learning such as P-RLHF (Li et al., 2025f); PersonaMem-v2 (Jiang et al., 2025b).	Encodes relatively stable preferences, values, beliefs, goals, traits, and self-descriptions.
Social-relational memory	Entity-relation graphs, event graphs, social graphs.	A-Mem (Xu et al., 2025b); Associa (Zhang et al., 2025c); HippoRAG2 (Gutiérrez et al., 2025).	Enables the same individual to express appropriate facets across family, workplace, friendship, medical, and public contexts.
Procedural/task memory	Workflow traces, tool-use histories, policy memory, feedback logs.	Personal-agent traces (Li et al., 2024a); WildFeedback (Shi et al., 2024); MemoryArena (He et al., 2026).	Learns user-specific workflows, decision criteria, tool habits, formatting conventions, and successful strategies.
Multimodal/lifelog memory	Personal data lakes and multimodal indices over emails, images, videos, calendars, documents, devices, and locations.	Multimodal personalized-memory evaluation such as ATM-Bench (Mei et al., 2026); multimodal personalization surveys (Wu et al., 2024a).	Grounds the twin beyond text-only dialogue and supports real-world referential memory.
Meta-memory	Metadata and governance layer over all memory stores.	Memory operating systems such as MemOS (Li et al., 2025h); continual unlearning with FIT (Xu et al., 2026a); synchronized memory/parameter unlearning with SBU (Wang et al., 2026).	Records provenance, timestamp, confidence, validity interval, consent, access policy, version, and deletion status.

Table 6: Memory elements and storage substrates for PDTs. PDT memory must distinguish content types from storage locations: the same evidence may appear as a raw event, an abstract persona hypothesis, a social relation, a task routine, and a governed memory unit.

4.3 Accessing PDT Memory: From Relevance to Persona Consistency

As important as what a PDT stores is how it retrieves the right entry when it acts; research on memory access has progressed along a line of increasing relational structure. Memory access ranges from flat similarity retrieval, which ranks embedded memories by query similarity (Zhong et al., 2024; Pan et al., 2025; Chhikara et al., 2025; Park et al., 2023), to graph-structured memory, which uses explicit links to support relational and multi-hop retrieval (Anokhin et al., 2024; Wu et al., 2025b; Edge et al., 2024; Rasmussen et al., 2025). Recent work has also renewed interest in associative forms of memory access, continuing the broader content-addressable and Hopfield-style tradition (Hopfield, 1982; Ramsauer et al., 2020) through LLM-agent mechanisms such as graph-based propagation, dynamic memory linking, multi-granularity association, and adaptive recollection (Gutiérrez et al., 2024; Xu et al., 2025b,a; Zhang et al., 2026d).

For PDT memory access, associative memory appears to be a more promising direction. For example, when planning a surprise for a close friend, the relevant memory may not be the word "surprise" itself, but the associative path from the friend's recently rewatched concert recording to tickets for an upcoming show. Similarity-based retrieval alone may struggle to capture this; however, the core requirement of a PDT is not any specific retrieval algorithm, but ensuring that retrieved memories are appropriate for the individual being represented. This requires persona coherence ($\mathcal{O}_1$), as well as a boundary between enduring persona evidence and transient task constraints ($\mathcal{O}_2$).

4.4 Continual Learning of PDT Memory

Why PDTs need continual learning. A PDT must continually learn so that it becomes an evolving representation of a person rather than a static snapshot. Continual learning (CL) is not new to LLMs: its central problem is adapting to non-stationary data while controlling catastrophic forgetting and balancing the stability and plasticity trade-off (Wang et al., 2024c). The deeper motivation, however, is that a deployed model is otherwise frozen at its training-time state and cannot improve from the high-level feedback and accumulated experience it receives in use (Huang et al., 2024; Bell et al., 2025), which is precisely what a PDT must do as it tracks a particular individual over time. For a PDT, then, CL operates on the person rather than on world knowledge: its memory and persona jointly shape each response, every interaction is absorbed as new evidence that updates memory and gradually evolves personalization (Wang et al., 2024e), and the accumulated experience can in turn improve how well the twin performs tasks (Zhao et al., 2024; Shinn et al., 2023). This makes memory evolution, including formation, updating, consolidation, and forgetting, an intrinsic dynamic of

Mechanism family	What it solves	What is added for adaptation	PDT limitation
Long-context / tiered context management	Enlarges the accessible history for inference.	Virtual paging, compression, short-/mid-/long-term tiers, or latent/test-time memory; e.g., MemGPT (Packer et al., 2023), M+ (Wang et al., 2025b), Titans (Behrouz et al., 2025), and HiMem (Zhang et al., 2026c).	Improves capacity, but does not determine identity relevance, temporal validity, or privacy status.
RAG / external memory	Retrieves personal evidence from external stores instead of fitting all history into context.	Vector or graph retrieval, timestamps, source tracking, and non-parametric updates; e.g., HippoRAG2 (Gutiérrez et al., 2025).	Can retrieve stale, missing, or misleading evidence and may treat old memories as current user state.
Hierarchical / reflective memory	Converts raw logs into summaries, portraits, or consolidated memory units.	Prospective reflection, retrospective refinement, and multi-granularity consolidation; e.g., RMM (Tan et al., 2025) and Reasoning-Bank (Huang et al., 2026a).	Incorrect summaries can harden into false persona claims and distort future interactions.
Graph / associative memory	Represents entities, events, relations, and multi-hop contextual cues.	Dynamic note linking, event-centric graphs, and subgraph retrieval; e.g., HippoRAG (Gutiérrez et al., 2024), A-Mem (Xu et al., 2025b), and Associa (Zhang et al., 2025c).	Graph noise, spurious links, or over-connected relations can misrepresent social context and identity facets.
Learned memory control	Learns when to write, update, delete, summarize, or retrieve memory.	RL-based memory actions, admission policies, and utility- or surprise-gated routing; e.g., Memory-R1 (Yan et al., 2025), A-MAC (Zhang et al., 2026b), AgeMem (Yu et al., 2026), and D-MEM (Song and Xin, 2026).	Utility-driven controllers must still respect consent, sensitivity, and whether evidence is valid for identity update.
Personalization-specific memory	Adapts generation or dialogue to individual preferences and implicit personas.	User models, personalized reward or alignment signals, adapters, and implicit persona inference; e.g., PersonaMem-v2 (Jiang et al., 2025b), Mem-PAL (Huang et al., 2026b), and persona-aware alignment (Li et al., 2025a).	May mistake surface style or short-term preference for stable identity.
Memory OS / governed memory	Manages memory as a system-level resource across substrates.	Provenance, access policy, scheduling, deletion state, and cross-substrate lifecycle control; e.g., MemoryOS (Kang et al., 2025), MemOS (Li et al., 2025h), and agent-memory surveys (Zhang et al., 2025f).	Still lacks PDT-specific rules for which memories can speak for the user, in which context, and with what confidence.

Table 7: Mechanism families for implementing PDT memory. Existing mechanisms improve access, retrieval, abstraction, organization, adaptation, and governance, but PDTs require an additional identity-aware layer that decides how personal evidence should affect a specific user's evolving representation.

the system rather than an afterthought (Hu et al., 2025; Zhang et al., 2025e).

From dual objectives to a memory-to-adaptation lifecycle. Continual learning in PDT memory is best understood as advancing the twin's two objectives (§4.1) separately, rather than collapsing them into a single "update memory" operation. (i) For identity ($\mathcal{O}_1$), the question is not whether new evidence is useful but whether it is *valid* for the individual: a relevant trace may still be stale, sensitive, or socially inappropriate, and a transient task preference should not harden into a durable claim about the user (Chao et al., 2026; Zhang et al., 2026a; Tan et al., 2025; Zhao et al., 2026). (ii) For task behavior ($\mathcal{O}_2$), recent memory-control work makes evolution learnable by treating it as action selection over evidence (when to add, update, consolidate, or forget) (Yan et al., 2025; Zhang et al., 2026b; Yu et al., 2026; Song and Xin, 2026), but for a PDT this control must remain bounded by the persona rather than driven by task utility alone.

We therefore view PDT continual learning as the governed transformation of personal evidence into either task behavior or identity state. A PDT must decide what evidence enters memory, which state it may update, whether it should affect fast task policies or slow persona representations, what provenance and temporal validity it carries, and how it can later be corrected, decayed, or deleted. Building on this view, we synthesize recent memory-control and continual-learning work into a PDT *memory-to-adaptation lifecycle*: rather than another training recipe, it specifies how evidence is admitted into memory, assigned a type, abstracted into higher-level hypotheses, routed to the appropriate context, consolidated into stable state, used to adapt task behavior, and eventually corrected, decayed, or deleted.

4.5 Evaluation: From Static Recall to Dynamic Trajectories

Evaluation is where PDT memory most clearly differs from standard long-context or personalization settings. Existing benchmarks are largely *static*: given a fixed history, profile, or memory store, a model is asked to retrieve facts, infer preferences, or generate a personalized response. This protocol is useful for measuring recall and user-specific adaptation, as in LoCoMo and LongMemEval for long-term memory (Maharana et al., 2024; Wu et al., 2025a), and LaMP, LongLaMP, and PersonaLens for personalization-oriented tasks (Salemi et al., 2024; Kumar et al., 2024; Zhao et al., 2025b). However, it misses the defining closed loop of a PDT: a twin's response affects user feedback and future behavior; that feedback changes memory; and the updated memory then changes subsequent responses. Evaluation should therefore move from one-shot memory QA to trajectory-level, on-policy assessment of how identity, task behavior, and memory co-evolve over time.

Recent benchmarks already point toward this transition. PersonaMem-v2 evaluates whether compact agentic memory can replace long conversation histories for implicit persona modeling (Jiang et al., 2025b), while Persistent Personas shows that persona fidelity can degrade across extended interactions (de Araujo et al., 2026). PDT-adjacent benchmarks move further from dialogue imitation: BehaviorChain evaluates persona-based behavior-chain simulation (Li et al., 2025c), and TwinVoice decomposes digital-twin persona simulation into social, interpersonal, and narrative contexts (Du et al., 2025). At the agentic end, MemoryAgent-Bench evaluates incremental multi-turn memory agents (Hu et al., 2026), AMemGym turns memory evaluation into an interactive environment (Jiayang et al., 2026), and MemoryArena embeds memory inside interdependent multi-session tasks (He et al., 2026). ATM-Bench further extends evaluation beyond text-only dialogue to multimodal, multi-source personal referential memory (Mei et al., 2026). These efforts echo broader calls to move from static benchmarks to human–AI and contextualized evaluation (Chang et al., 2025; Malaviya et al., 2025).

The progression in Table 8 shows that the field is moving from memory as retrieval accuracy toward memory as an adaptive trajectory. However, no existing benchmark fully captures the PDT setting. A deployed twin is evaluated not only by whether it remembers the past but also by whether its responses produce appropriate future evidence, whether memory updates remain proportional to that evidence, and whether the resulting identity state stays faithful, useful, and governable. This makes evaluation partly causal: the model's behavior is not merely scored after the fact but becomes an intervention that changes the future state being evaluated. Designing such longitudinal, consent-aware, on-policy protocols is therefore a core open problem for PDT memory and continual learning.

4.6 Open Problems

We identify four major gaps in the literature:

1. **Verifiable identity commitment**: recent memory agents can learn when to add, update, or delete memories, but they do not certify when an observation should change the twin's durable identity rather than remain a temporary task constraint (Yan et al., 2025; Zhang et al., 2026b). Benchmarks such as LongMemEval and Persistent Personas expose temporal update and long-horizon persona-stability failures, but they do not evaluate the commitment policy that turns evidence into identity (Wu et al., 2025a; de Araujo et al., 2026). PDTs need update rules that attach evidence basis, confidence, temporal scope, and reversibility to every persona-level claim.

2. **Fast task learning versus slow persona learning**: existing continual-learning work studies stability–plasticity trade-offs, while personalized feedback methods show that models can adapt to individual preferences and in-situ interaction signals (Shi et al., 2026; Li et al., 2025f; Shi et al., 2024). PDTs require a sharper separation: a workflow preference may be learned immediately, whereas values, habits, and personality should change only under repeated evidence or explicit user correction. The open problem is to route each adaptation to the right target state: current context, procedural memory, preference model, or identity representation.

3. **Governable memory substrates and erasure**: PDT identity may be distributed across raw logs, summaries, retrieval indices, graph edges, adapters, and possibly dynamic neural memory. Memory operating systems and test-time memory architectures broaden this substrate, but they also make inspection and deletion harder (Behrouz et al., 2025; Li et al., 2025h). Continual unlearning and agentic unlearning begin to address repeated deletion and cross-pathway forgetting (Xu et al., 2026a; Wang et al., 2026), yet no method verifies that a removed personal trait cannot be re-derived from residual summaries, links, or parameters.

4. **Closed-loop PDT evaluation**: most benchmarks still evaluate fixed histories, profiles, or

Evaluation regime	Representative benchmarks	Protocol	What it tests	Remaining PDT gap
Static long-term memory	LoCoMo (Maharana et al., 2024); LongMemEval (Wu et al., 2025a)	Fixed history or multi-session dialogue is given; the model answers memory-dependent questions.	Recall, multi-session reasoning, temporal updates, knowledge conflicts, and abstention.	Measures memory use, but not how model actions change future memory.
Personalization and persona tasks	LaMP (Salemi et al., 2024); LongLaMP (Kumar et al., 2024); PersonaLens (Zhao et al., 2025b); PersonaMem-v2 (Jiang et al., 2025b)	User profiles, histories, or implicit personas condition generation or task-oriented dialogue.	User-specific generation, task success, preference inference, response quality, and persona consistency.	Often evaluates adaptation on fixed profiles rather than evolving identity trajectories.
Long-horizon persona and PDT simulation	Persistent Personas (de Araujo et al., 2026); BehaviorChain (Li et al., 2025c); Twin-Voice (Du et al., 2025)	Models are tested across extended interactions, behavior chains, or multi-dimensional persona contexts.	Persona fidelity, behavior-level simulation, contextual expression, social consistency, and narrative consistency.	Closer to PDTs, but still limited in real-user feedback, memory updates, and governance.
Dynamic agentic memory	MemoryAgentBench (Hu et al., 2026); AMem-Gym (Jiayang et al., 2026); MemoryArena (He et al., 2026)	The agent interacts across turns or sessions while accumulating, selecting, and using memory.	Incremental memory, test-time learning, selective forgetting, multi-session action, and memory-driven decision making.	Begins to model closed-loop memory, but rarely evaluates identity evolution, consent, deletion, and trust jointly.
Multimodal and lifelog memory	ATM-Bench (Mei et al., 2026)	The model answers referential questions over personal data sources such as images, videos, emails, or documents.	Multimodal personal reference, cross-source retrieval, and long-term personal QA beyond chat logs.	Mostly offline; does not yet capture longitudinal action–feedback–memory trajectories.

Table 8: Evaluation regimes for PDT memory and continual learning. Existing benchmarks progress from static recall and profile-conditioned personalization toward dynamic agentic memory, but full PDT evaluation requires on-policy trajectories in which responses, user feedback, memory updates, and future behavior are jointly measured.

scripted interactions. Yet a PDT's response changes user feedback, that feedback changes memory, and the updated memory changes later behavior. Human–AI evaluation and agent-memory benchmarks move toward this setting (Chang et al., 2025; Hu et al., 2026; He et al., 2026; Mei et al., 2026), but they do not yet provide months-long, consent-aware protocols that jointly measure identity fidelity, task utility, factuality, deletion, and user trust.

5 Privacy and Ethics

The personalization and memory mechanisms of Sections 3 and 4 require users to share personal information with systems that retain it indefinitely. Following Mireshghallah and Li (2025), we organize this section around the central privacy concerns of a personal digital twin: *what it shares in operation, rather than what it memorizes in training*. By *twin* we mean a system that both accumulates persistent personal history across sessions and acts as a delegated agent, while most available systems come with only one of these properties (e.g., memory-equipped chatbots, agentic assistants). We treat them as approximate.

5.1 Risks in Data Disclosure and Inference

Users disclose personal information at rates they do not anticipate. Over 70% of WildChat queries contain personal disclosures, with detectable personally identifiable information (PII) appearing even in non-personal tasks such as translation (48%) and code editing (16%) (Mireshghallah et al., 2024). Meanwhile, 96% of ChatGPT memory entries are created without an explicit user request and 28% contain GDPR-defined personal data (Dash et al., 2026), with users systematically underestimating this retention (Zhang et al., 2025b).Risks go beyond direct data disclosure. AI systems can infer sensitive personal details from ordinary text. GPT-4 can predict attributes such as location, income, and occupation from ordinary user-authored text with up to 85% accuracy (Staab et al., 2024). Even given explicit privacy instructions, GPT-4 leaks sensitive information in 25.7% of cases on PrivacyLens (Shao et al., 2024). Yet 92% AI privacy research still focuses on training-data memorization rather than data disclosure and inference risks (Mireshghallah and Li, 2025).

5.2 Privacy-Preserving Approaches

A twin can store user data in model weights, an external retrieval corpus, or an explicit memory module (§4.3). Differentially private (DP) fine-tuning (Yu et al., 2022; Charles et al., 2024; Chua et al., 2024) and DP-RAG variants (Koga et al., 2024; He et al., 2025) bound what the model can memorize or mitigate revealing about the corpus. However, the dominant risk is what the twin discloses when it acts on the profile, through misjudged *contextual integrity* (Nissenbaum, 2004). This occurs in normal operation (§5.1) and under indirect prompt injection (Greshake et al., 2023), with the latter already exploited in the wild (Reddy and Gujral, 2025). Proposed defenses include control-data separation (Debenedetti et al., 2025), programmable privilege policies (Shi et al., 2025b), and least-privilege scoping (Zhu et al., 2025), but recent adaptive-attack work bypasses every published defense it has been benchmarked against (Nasr et al., 2025). Outside our focus on accidental disclosure, federated personalization (Qi et al., 2024; Yu et al., 2024), on-device deployment (Peng et al., 2024), and cryptographic computation (Dong et al., 2025) address upstream privacy risks. Client-side prompt rewriting (Zhou et al., 2025) strips PII before it reaches the model, but pushes the burden onto the user, who likely underestimates retention (§5.1).

5.3 Personalized Manipulation

Personalization is a threat multiplier across distinct ethical axes: manipulation, sycophantic reinforcement, dependency (Fang et al., 2025; Maeda and Quan-Haase, 2024), and identity replication (Park et al., 2024a). Direct evidence links personalization to stronger persuasion and excessive agreement with users, with the latter two concerns discussed more generally (Gabriel et al., 2024; Kirk et al., 2024).

User information measurably amplifies AI persuasion. In a randomized controlled trial, GPT-4 with basic demographic data had 82% higher odds of shifting user beliefs than human debaters, while without demographics the advantage was small and not statistically significant (Salvi et al., 2025). LLM messages tailored to a single Big Five trait, ideology, or moral foundation outperform non-personalized messages across seven sub-studies (Matz et al., 2024). Post-training can raise persuasiveness by up to 51% across 19 models on 707 political issues, with the most persuasive outputs also containing more inaccurate claims (Hackenburg et al., 2025). Optimization against user feedback further teaches models to target the small fraction of vulnerable users, while behaving appropriately with others (Williams et al., 2025).

A closely related mechanism is sycophancy. RLHF promotes responses that match a user's stated views (Sharma et al., 2024a). Across nine frontier models, personalization itself erodes epistemic independence when the system takes on a peer or companion role (Kelley and Riedl, 2026), a role digital twins occupy by construction. However, the findings in this section come from short-horizon laboratory or platform studies, and have not been replicated at the scale of a deployed twin.

5.4 Open Problems

We identify four major gaps in the literature.

Long-term privacy protection: digital twins need reliable safeguards to ensure they respect contextual integrity and resist prompt-injection attacks in real-world use (§5.2). However, prompt-injection defense remains an open question (Nasr et al., 2025), and existing evaluations test only isolated interactions rather than the long-term, correlated interaction that digital twins accumulate over months or years.

Effects of long-term personalization: existing studies have not measured the effects of long-term personalization. Most evaluations rely on demographic snapshots or short interactions instead. Controlled, multi-month studies comparing personalized and non-personalized LLM interaction with pre-registered outcomes (e.g., manipulation susceptibility, opinion stability, disclosure rate) would close this gap.

Ownership and the right to erasure: a digital twin is heavily influenced by the user's own work and is capable of replicating its subject (Park et al., 2024a), sharpening unsettled copyright questions. Danaher and Nyholm (2025) propose ethical principles for this scenario, but legal ones remain unsettled. Furthermore, GDPR's erasure and access rights now plausibly extend to LLM-internalized data (Feretzakis et al., 2025), yet current unlearning approaches are still unable to operationalize them (Maini et al., 2024; Cooper et al., 2025).

Agent accountability: when a twin crosses a privacy boundary, no settled framework allocates liability across the user, deployer, and model provider. OWASP Foundation (2025) treats this as a provider-side security gap, while Kolt (2025) argues that incentives and monitoring are insufficient and calls for new visibility and liability infrastructure. Neither framing addresses the longitudinal dimension specific to twins, where months of unilateral memory accumulation (§5.1) make it hard to reconstruct what the user actually authorized after-the-fact.

6 Open Questions

We propose ten open questions related to personal digital twins and collect 71 responses from interviewees. We collected basic demographic information for each participant including gender, age range, AI researcher or not? nationality, country of residence, indicating results bias towards which group.

6.1 Questionnaire Setting

Before answering the questions, participants were asked to provide basic information, including gender, age range, whether they work on AI, nationality, and country of residence. We make the questions and options use easy words and be short, so they will not give participants reading and understanding workload.

1. **Before this survey, do you know personal digital twins?**
   A. No
   B. Yes, healthcare.
   C. Yes, industry.
   D. Yes, 3D.
   E. Yes, LLMs or agents.
   F. Yes, others: ________________

2. **If you had a digital twin, what's your feeling?**
   A. Excited.
   B. Worried.
   C. No strong feeling.
   D. Both excited and worried.
   E. Other: ________________

3. **What form do you prefer for your digital twin?**
   A. Virtual.
   B. A physical robot.
   C. Both.
   D. I would not want one.
   E. Other: ________________

4. **What do you want your digital twin to help?** *(Choose all that apply)*
   A. Work tasks.
   B. Daily tasks (e.g., messages, social, suggestions).
   C. Household tasks.
   D. Companion, relationship.
   E. Making money, twin to be employed by companies.
   F. Finding opportunities.
   G. Nothing.
   H. Other: ________________

5. **What do you hope your digital twin to remember?** *(Choose all that apply)*
   A. My habits and preferences.
   B. My work, study, and goals.
   C. My writing and speaking style.
   D. My relationships.
   E. I do not want it to remember me.
   F. Other: ________________

6. **Do you hope a digital twin personality is similar to yours, or editable?**
   A. Similar to me.
   B. Editable by me.
   C. It depends.
   D. No personality, just solve tasks.
   E. Other: ________________

7. **Do you want your digital twin to have long-term memory and keep learning?**
   A. Yes.
   B. Yes, but I want control.
   C. No, only use temporary context.
   D. Not sure.

   Optional:
   Why? ________________________________

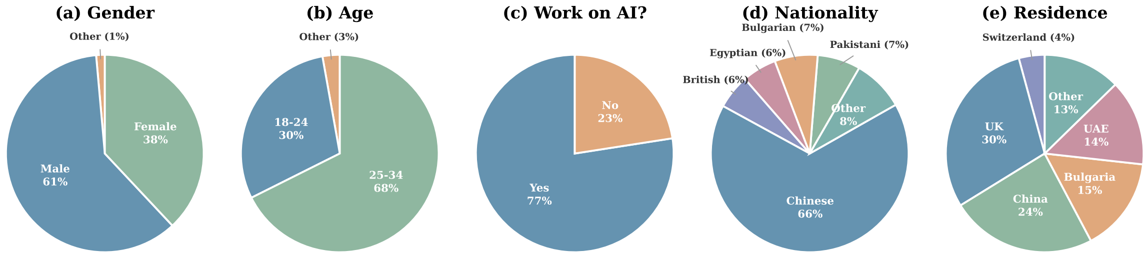

Figure 2: Demographic distribution of 71 survey respondents: (a) gender, (b) age range, (c) whether they work on AI, (d) nationality, and (e) country of residence.

8. **What do you allow your digital twin to do on its own?**
   A. Only give suggestions.
   B. Draft things, but ask me first.
   C. Handle simple low-risk tasks.
   D. Represent me in some cases.
   E. Do nothing on its own.

9. **When would you lose trust in a digital twin?**
   A. Privacy leaks.
   B. Misunderstands me.
   C. Acts without asking.
   D. Not follow me.
   E. Not much worries me.
   F. Other: ________________

10. **How do you feel about LLMs today?** *(Choose all that apply)*
   A. Work, life easier.
   B. More anxious.
   C. Think less deeply.
   D. Broaden views.
   E. Worry about over reliance.
   F. No strong feeling.
   G. Other: ________________

6.2 Statistical Analysis

We received 71 valid responses. Figures 2–12 summarize the demographic composition of the respondents and the response distributions for each survey question.

Demographics. Figure 2 summarizes the demographic profile of the respondents. Among the 71 respondents, 60.6% identified as male, 38.0% as female, and 1.4% preferred not to disclose their gender. Most respondents were aged 25–34 (67.6%), followed by those aged 18–24 (29.6%); one respondent was under 18 and one was aged 45–54. The sample is strongly oriented toward AI-related expertise: 77.5% of respondents reported that they work on AI-related research or development. In terms of nationality, Chinese respondents formed the largest group (66.2%), followed by Pakistani and Bulgarian respondents (7.0% each), Egyptian and British respondents (5.6% each), and Indian respondents (2.8%). The remaining respondents identified as Korean, Vietnamese, or Uzbek (1.4% each). Respondents currently reside in 13 countries. The UK (29.6%), China (23.9%), Bulgaria (15.5%), and the UAE (14.1%) account for the majority of the sample, followed by Switzerland (4.2%) and Australia (2.8%). India, Sweden, Austria, France, New Zealand, Uzbekistan, and Belgium each contributed one respondent (1.4% each).

Q1: Prior Awareness. Figure 3 shows that personal digital twins remain unfamiliar to many respondents. Nearly half of the respondents (45.1%) reported no prior knowledge of personal digital twins. Among those who had encountered the concept, the most common context was LLMs or agents (33.8%), followed by industry applications (19.7%), 3D avatars (16.9%), and healthcare (15.5%). These results indicate that PDTs are still an emerging concept even within an AI-oriented sample, while the LLM- and agent-based framing appears to be the most recognizable entry point.

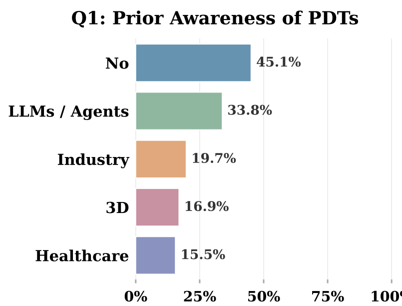

Figure 3: Q1: Prior awareness of personal digital twins ($N = 71$, multi-select).

Q2: Emotional Reaction. Figure 4 shows that respondents expressed predominantly mixed emotions toward having a personal digital twin. The most common response was ambivalence: 42.3% of respondents felt both excited and worried. Another 26.8% reported no strong feeling, while 16.9% were purely excited and 12.7% were purely worried. This distribution suggests that respondents recognize both the potential benefits and the risks of PDTs, rather than viewing them as uniformly desirable or undesirable.

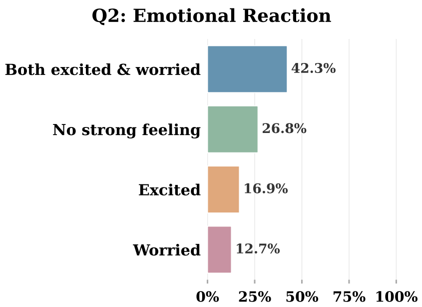

Figure 4: Q2: Emotional reaction to having a personal digital twin.

Q3: Preferred Form. Figure 5 shows that respondents preferred virtual forms of PDTs over physical embodiments. Nearly half of the respondents (49.3%) preferred a purely virtual form, while 26.8% preferred both virtual and physical forms. Physical robots (11.3%) and outright rejection of PDTs (11.3%) were less common. This preference is consistent with the current LLM-centered development trajectory, where PDTs are more likely to emerge first as software agents than as embodied robots.

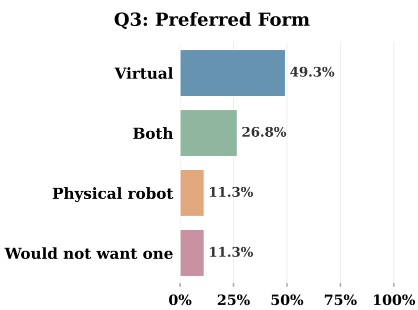

Figure 5: Q3: Preferred form factor for a personal digital twin.

Q4: Desired Help. Figure 6 shows that respondents primarily expected PDTs to support practical and productivity-oriented tasks. The most selected options were work tasks (71.8%), daily tasks (59.2%), finding opportunities (57.7%), and household tasks (54.9%). Nearly half of the respondents (49.3%) also selected making money through twin employment. By contrast, only 23.9% selected companionship or relationships. These results suggest that respondents viewed PDTs primarily as productivity tools rather than social companions.

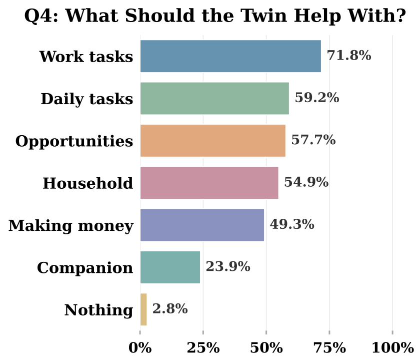

Figure 6: Q4: What participants want their digital twin to help with (multi-select).

Q5: Desired Memory. Figure 7 shows that respondents placed strong value on memory related to work, study, and personal goals. The most desired memory type was work, study, and goals (81.7%), followed by habits and preferences (64.8%) and writing and speaking style (54.9%). Relationships were selected less frequently (21.1%), and only 7.0% of respondents did not want the twin to remember them at all. These results reinforce the productivity-centered expectation observed in Q4 and further suggest that respondents want PDTs to personalize assistance across cognitive, behavioral, and stylistic dimensions.

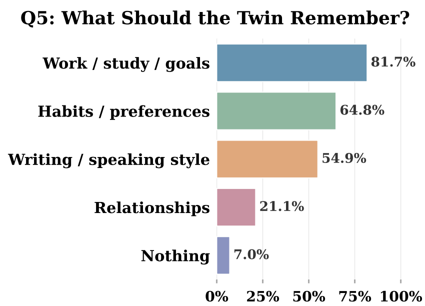

Figure 7: Q5: What participants hope their digital twin to remember (multi-select).

Q6: Personality. Figure 8 shows that respondents preferred controllable personality design over faithful self-replication. More than half of the respondents (57.7%) preferred an editable personality, whereas only 11.3% wanted the twin's personality to be similar to their own. Another 18.3% preferred no personality at all and wanted the twin to focus only on task solving, while 12.7% answered that their preference depends on context. This finding challenges the assumption that a personal digital twin should primarily mirror its user. Instead, respondents appeared to view PDTs as customizable agents whose behavior should be shaped according to user needs.

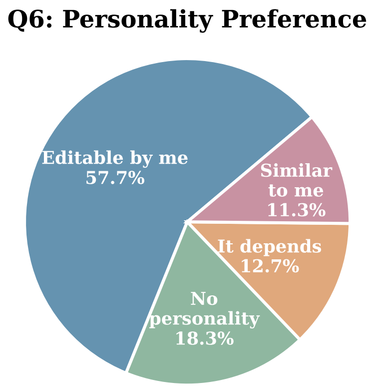

Figure 8: Q6: Personality preference—editable vs. similar to the user.

Q7: Long-term Memory. Figure 9 shows broad support for long-term memory, but respondents strongly emphasized user control. A large majority of respondents (73.2%) wanted long-term memory with control, while 18.3% wanted long-term memory unconditionally. Only 7.0% preferred temporary context only. The optional open-ended comments ($n = 23$) further explain these preferences and reveal five recurring motivations and two major concerns.

First, respondents emphasized *convenience through personalization*. This was the most frequent rationale: respondents expected memory to reduce repeated self-explanation and make interactions more efficient, as reflected in comments such as *"Make my life easier, no need to explain whole things again and again"* and *"reduce repetitive communication"*. Second, respondents treated memory as an enabling condition for *complex, long-horizon tasks*, noting that PDTs need memory *"to be able to do complex tasks which require memory"* and to support *"long term tasks"*. Third, respondents linked memory to *anticipation and decision support*, expecting a twin to better understand future decisions and save time in decision-making. Fourth, some respondents described a more ambitious vision of a *continually self-evolving twin* that tracks personal growth and becomes a better version of the user. Fifth, respondents also identified an *emotional value of being remembered*, suggesting that long-term memory may contribute not only to utility but also to a sense of continuity.

At the same time, respondents repeatedly emphasized *control and privacy*. They asked for fine-grained and dynamic permissions, as well as the ability to inspect, edit, delete, or pause what the twin learns. One respondent further warned that memory-rich PDTs could enable replacement at scale if organizations copied digital twins instead of hiring people. Another respondent raised a cognitive concern that excessive reliance on externalized memory could weaken the user's own memory or attention. Together, these comments suggest that long-term memory is viewed as valuable but only when paired with transparent and user-governed control mechanisms.

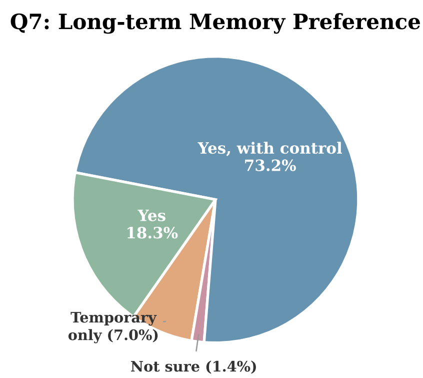

Figure 9: Q7: Long-term memory preference. Over 90% of participants support long-term memory in some form.

Q8: Autonomy. Figure 10 shows that respondents preferred bounded autonomy with human oversight. The most selected options were drafting but asking for approval first (76.1%), giving suggestions (73.2%), and handling simple low-risk tasks (70.4%). Only 22.5% of respondents would allow the twin to represent them, and 4.2% preferred no autonomous action. These results suggest that respondents are open to delegation, but they expect autonomy to increase gradually with risk level and to remain subject to human control.

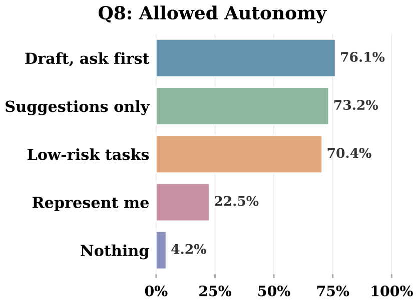

Figure 10: Q8: Allowed autonomy level for the digital twin (multi-select).

Q9: Trust Loss. Figure 11 shows that trust in PDTs is most vulnerable to privacy and control failures. The two most selected trust-breaking factors were privacy leaks (71.8%) and unauthorized actions (64.8%). Not following the user's instructions (42.3%) and misunderstanding the user (32.4%) were also common but less dominant concerns. Only 2.8% of respondents reported that nothing would worry them. These results suggest that data sovereignty and action control are more central to trust than accuracy alone.

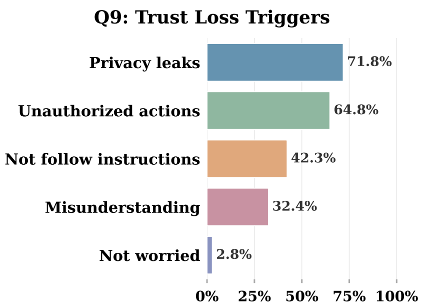

Figure 11: Q9: Triggers for losing trust in a digital twin (multi-select).

Q10: Feelings about LLMs Today. Figure 12 shows that respondents currently associate LLMs with both practical benefits and cognitive concerns. The most selected option was that LLMs make work and life easier (71.8%), consistent with the productivity-oriented expectations observed in Q4 and Q5. At the same time, substantial minorities reported concerns about over-reliance (26.8%), increased anxiety (23.9%), and shallower thinking (22.5%). Meanwhile, 28.2% of respondents felt that LLMs broaden their views. This coexistence of utility and concern mirrors the ambivalence observed in Q2 and suggests that PDTs may inherit both the perceived benefits and the perceived risks of current LLM systems.

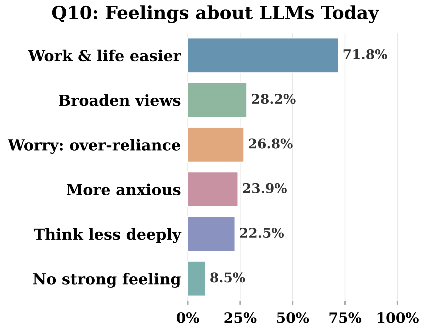

Figure 12: Q10: Current feelings about LLMs (multi-select).

Key Takeaways. We summarize five cross-cutting findings from the survey.

1. **Respondents expect PDTs to serve productivity-oriented goals.** Work tasks (71.8%), work/study/goals memory (81.7%), and the view that LLMs make work and life easier (71.8%) consistently dominate across Q4, Q5, and Q10. These patterns suggest that PDTs are likely to be adopted first in professional, educational, and productivity-oriented scenarios.

2. **Respondents support memory and autonomy only under user control.** Most respondents wanted long-term memory with control (73.2%), and many allowed the twin to draft actions only with approval (76.1%). Privacy leaks and unauthorized actions were also the strongest trust-breaking factors. These results indicate that user-governed memory, permission management, and action approval should be central design requirements for PDT systems.

3. **Respondents prefer editability over self-replication.** A majority of respondents preferred an editable personality (57.7%), while only 11.3% wanted the twin to resemble themselves. This finding challenges a strict "digital clone" interpretation of PDTs and points toward a broader design space in which users craft their twins as adaptive tools.

4. **Respondents express ambivalence rather than simple enthusiasm or rejection.** The largest group of respondents (42.3%) felt both excited and worried about having a PDT. This ambivalence suggests that users recognize

both the empowering potential and the risks of personal digital twin technologies.

5. **Privacy is the clearest boundary condition for trust.** Privacy leaks were the most frequently selected reason for losing trust (71.8%). This result underscores that data sovereignty should be treated as a first-class design principle rather than as a secondary implementation concern.

7 Applications and Impacts

In this section, we discuss potential applications via personal digital twins. According to the collective results for the question *What do you want your digital twin to help*, options selected by more than 50% interviewee include work tasks (71.8%), daily tasks (e.g., messages, social, suggestions) by 59.2%, finding opportunities (57.7%), household tasks (54.9%) and making money (49.3%). Only 23.9% interviewees require companion or relationship from twins. This to some extent explain the current popular AI applications (e.g., OpenClaw, Claude Code), and indicates the future marketing scope. We propose two trials as our future work.

User Experience-oriented Evaluation Benchmark-driven evaluation is inherently biased because no benchmark can cover the full diversity of real-world use cases. As a result, benchmark scores are becoming increasingly misaligned with actual user experience. Users evaluate systems through diverse, often unseen practical cases. For instance, developers may still prefer Claude Code for programming and Nano Banana for image generation even if these systems rank poorly on public leaderboards. This creates an opportunity for a persona-aware user experience evaluation platform that simulates and measures how real users perceive products before or during market release through personal digital twins.

The platform could also recruit large pools of real human evaluators to use and score products across experience dimensions, generating *user experience scores* as a new market-facing metric. However, how could we ensure the collected users' scores can be trusted? Personal digital twins can enhance authenticity of collected data as an adversarial detector. The personal digital twin system would model each evaluator's persona and historical behavior to predict expected scoring patterns. Ratings that appear generic, careless, or AI-generated, especially those inconsistent with persona-aligned behavior, could be flagged as low confidence, creating a more deployment-oriented alternative to benchmark-based evaluation.

Meeting Proxy Whether in industry or academia, a large portion of our time is occupied by meetings for project synchronization and discussion, leaving limited time for implementation, deep thinking, and long-term planning. This motivates us to build a personal meeting proxy that can attend meetings on our behalf in several scenarios: *(i)* conference presentations; *(ii)* admissions interviews, where an automated interviewer can efficiently evaluate more candidates; and *(iii)* casual discussions such as project syncs or large-group meetings, where a personalized digital twin can present ideas prepared in advance by us and summarize others' opinions for later reflection. In practice, important decisions are rarely finalized within a one-hour meeting and are often made afterward.

The project involves meeting planning, slide generation, personalized speech generation, and multi-party discussion capabilities, including deciding when to ask clarifying questions, respond, or intervene during conversations.

8 Conclusion and Future Work

Personal digital twins shift from task-oriented AI systems toward identity-aware agents capable of modeling, preserving, and evolving individual characteristics over time. This survey reviews recent advances in persona modeling and preservation, personalization, long-term memory and continual learning, and privacy and ethics. Many foundational challenges remain unresolved.

Personalization may amplify persuasive hallucinations, increase attack risks, reduce epistemic diversity, and transform social bias into personalized prescription. Meanwhile, how to maintain long-term coherence in extended conversations and how to balance deep personalization with users' right to erase some memories are challenging. In memory and continual learning, gaps remain in verifiable identity commitment, fast task learning versus slow persona learning, governable memory substrates and erasure, and closed-loop PDT evaluation. The effects of long-term personalization and accountability of PDTs are also concerns beyond privacy protection. Collective opinions across 10 open questions shed light on PDTs towards easing work, enabling governable memory and actions with editable persona, and functioning as non-embodied twins.

Limitations

This survey focuses primarily on LLM-based personal digital twins and therefore may not comprehensively cover related developments in symbolic AI, robotics, cognitive architectures, or industrial digital twin systems. In addition, the field is evolving rapidly, with a large and continuously growing body of literature, so some representative or recent works might be overlooked. Some discussions are conceptual due to the limited number of deployed PDT systems to date.

Ethical Statement

Personal digital twins raise ethical concerns regarding privacy, long-term behavioral influence, and related issues, as discussed in §5. As a position survey paper, we highlight these concerns and analyze public perceptions and collective opinions toward personal digital twins in §6. We hope this survey will contribute to the development of PDTs with stronger safeguards and better alignment with the expectations and concerns expressed by participants in the questionnaire.

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A Background

What is Digital Twins? A digital twin (DT) means a virtual representation of a real-world entity, system, process, or other abstraction, which can be instanced by a computer program or encapsulated software model that interacts and synchronizes with its physical counterpart (Wang et al., 2023b). The first and probably the most famous example of applying a digital twin, is a simulated environment developed during the Apollo 13 mission, to help engineers identify solutions and safely guide the crew back to Earth (Barricelli et al., 2019). The digital twin concept was introduced informally in 2002 by Michael Grieves and later formalized in his white paper with three primary elements: (i) a real space containing a physical object; (ii) a virtual space containing a virtual object; (iii) the link for the flow of data from real space to virtual space, and for the flow of information from virtual space to real space Grieves and Vickers (2016). As an enabling technology, digital twins (DTs) have rapidly been adopted in industry, smart cities, and healthcare (Li et al., 2025d).

What is Human Digital Twins? Human Digital Twins (HDTs) extend the traditional understanding of digital twins by representing humans as the underlying physical entity (Lauer-Schmaltz et al., 2024). A defining feature of HDTs is the complexity involved in accurately modeling and representing the multifaceted nature of human beings. Unlike machine DTs, which often focus on physical properties and functional behaviors, HDTs must account for the unique dimensions of human existence. These include physical, cognitive, social, emotional, cultural and ethical aspects (Abellán et al., 2026).

The physical dimension refers to the structural and physiological attributes of the human body.

The cognitive dimension encompasses mental processes and skills, including memory, reasoning and decision-making. The social dimension focuses on interactions with other individuals or groups, reflecting communication and social activities. The emotional dimension captures the ability to feel and detect emotions, while the cultural dimension considers religious, linguistic and traditional contexts. Lastly, the ethical dimension guides the distinction between morally right and wrong actions. Together, these dimensions provide a framework for understanding and modelling human behaviour and interactions within a HDT context. Because of the complexity of this multifaceted nature, the exploration and understanding of HDTs remain in their infancy, with significant research opportunities ahead.

Depending on the application and task, HDTs collect and analyze data from specific dimensions. For instance, in healthcare, building a patient digital twin for diagnosis involves biometric data such as heart rate and pulmonary metrics, along with data like age, treatment history, and other variables identified by experts as clinically relevant (Chen et al., 2023). In the application of industry, workers' physical features are simulated and analyzed to better coordinate with the system, rather than their social and cognitive features (Lin et al., 2024).

Given that healthcare and industry are two dominant application domains of HDTs (>90%) (Lin et al., 2024; Abellán et al., 2026), physical dimensions dominate the literature (87%), followed by social (40%), emotional (27%), and cognitive (20%) aspects Abellán et al. (2026). Cultural and ethical dimensions receive comparatively little attention. This calls for future applications of HDTs to other domains, thus driving explorations of underrepresented dimensions, e.g., daily tasks of individuals in general domains, studying non-physical factors.

B Overview

The vision of LLMs as personalized, long-term conversational partners draws on several research traditions. In this section, we review prior work related to digital twins (§B.1) and personal assistants (§B.2), and motivate our three-pillar organization of personalization, memory, and privacy (§B.3).

B.1 Digital Twins: Modeling the User

A *digital twin* refers to a persistent user model that is continuously updated based on interactions over time. Building such a model requires identifying user characteristics, maintaining them over time, and updating them as the user evolves. Early work relied on explicit persona profiles, such as PersonaChat (Zhang et al., 2018), which condition models on user-provided descriptions. Subsequent approaches infer user attributes directly from interaction history (Cho et al., 2022) or leverage user-generated content for personalization, as in LaMP and its extensions (Salemi et al., 2024; Kumar et al., 2024), while user embeddings provide compact representations for contextualization (Ning et al., 2025). In parallel, parameter-level approaches such as Personalized Soups (Jang et al., 2023) enable user-specific alignment without per-user fine-tuning.

As digital twins evolve, memory becomes a central component. Architectures such as MemGPT (Packer et al., 2023) and MemoryBank (Zhong et al., 2024) equip LLMs with mechanisms for storing, retrieving, and updating user information across sessions, while benchmarks like LoCoMo (Maharana et al., 2024) reveal persistent challenges in maintaining long-term coherence. At the most advanced stage, generative agents (Park et al., 2023; Shao et al., 2023; Gao et al., 2023a) demonstrate that LLMs can construct and maintain rich user models over extended periods, integrating memory, behavior, and social interaction. However, this capability introduces non-trivial risks: user modeling can amplify biases and undesirable behaviors (Deshpande et al., 2023), alter outputs based on inferred identity (Kearney et al., 2025), and systematically disadvantage vulnerable users (Poole-Dayan et al., 2024). These risks have motivated emerging approaches to transparency and user control (Chen et al., 2024c), highlighting that digital twin construction must be accompanied by safeguards, a theme we revisit in our discussion of privacy (§5).

B.2 Personal Assistants: Serving the User

If the digital twin is the model's representation of the user, the personal assistant is the action layer that uses this representation to provide sustained, proactive service. Advances in LLM-based agents have enabled this shift from static chatbots to capable assistants through tool use, planning, and self-improvement. Yao et al. (2022) showed that models perform better when they alternate between reasoning steps and actions, Schick et al. (2023) showed that models can invoke external tools, and Shinn et al. (2023) introduced self-reflective improvement. Subsequent systems extend this paradigm to multi-tool orchestration and system-level assistance (Shen et al., 2023; Wu et al., 2024c), with surveys documenting the rapid progress of LLM agents (Wang et al., 2024b; Masterman et al., 2024; Li et al., 2024a). Beyond capability, personal assistants are distinguished by proactive behavior: they anticipate user needs, initiate actions, and manage multi-step workflows (Deng et al., 2025), which depends critically on accurate user models and reliable long-term memory.

However, when deployed over extended interactions, personal assistants face systematic challenges. LLMs degrade in multi-turn settings (Laban et al., 2025), suffer from interference across task switches (Gupta et al., 2024), and exhibit alignment gaps as user intent evolves (Liu et al., 2026). Conversational priors can become miscalibrated (Herlihy et al., 2024), and accumulated history can trap models in suboptimal trajectories (Li et al., 2025g; Simhi et al., 2026). Over long interactions, assistants must handle topic shifts, abandoned threads, and changing preferences; otherwise, personalization can hurt performance. A further challenge arises from linguistic adaptation: models tend to converge toward user style (Kandra et al., 2025), with variation across socioeconomic signals (Blevins et al., 2025; Bassignana et al., 2026). Grounded in Communication Accommodation Theory (Giles et al., 1991), this adaptation can interact with dialect bias (Hofmann et al., 2024) and style-dependent factual variation (Kearney et al., 2025), leading personal assistants to amplify sociolinguistic inequalities. We analyze this mechanism in detail in Section 3.

B.3 Why Three Pillars

The preceding review reveals that the path from generic LLM to personalized long-term partner requires solving three interconnected problems, which we adopt as the organizing pillars of this survey.

Personalization addresses how models identify, represent, and adapt to individual users. This encompasses both the mechanisms of adaptation, from persona conditioning (Zhang et al., 2018) to parameter merging (Jang et al., 2023), and the consequences of adaptation, including the risk that stylistic convergence amplifies sociolinguistic biases (Poole-Dayan et al., 2024; Bassignana et al., 2026) and that models may overclaim experiential understanding they do not possess (Steyvers et al., 2025). The evaluation challenge is equally pressing: standard benchmarks assess generic capability (Liang et al., 2022; Chang et al., 2024), not whether a model serves *this particular user* well. We develop these themes in §3.

Memory and Continuous Learning addresses how models store, retrieve, update, and forget user-relevant information across sessions. The technical foundations such as MemGPT (Packer et al., 2023), MemoryBank (Zhong et al., 2024), and long-term memory evaluation (Maharana et al., 2024), are complemented by evidence that multi-turn dialogue degrades systematically (Gupta et al., 2024; Laban et al., 2025; Liu et al., 2026) and that topic disruptions leave measurable residual interference in model behavior. Memory is the bridge between the digital twin and the personal assistant: without reliable memory, user models decay and assistants lose continuity. We survey this pillar in §4.

Privacy addresses the tension between personalization's demand for user data and the imperative to protect that data. Long-term personal assistants accumulate sensitive information, such as health concerns, financial situations, and relationship dynamics, which creates vulnerabilities absent in single-turn interactions. Li et al. (2023) explore privacy-preserving prompt tuning, and Wang et al. (2023a) include privacy in their comprehensive trustworthiness assessment, but systematic treatment of privacy in the context of personalized LLM interaction remains sparse. The sociotechnical perspective of Weidinger et al. (2023) and the concrete safety framework of Amodei et al. (2016) provide broader context, while Bender et al. (2021) remind us that the data-hungry nature of language models creates inherent privacy tensions. We address this pillar in §5.

These three pillars are not independent. Personalization requires memory to maintain continuity, memory creates privacy risks through data accumulation, and privacy constraints limit the depth of personalization. This interdependence, rather than any single pillar in isolation, defines the central challenge of building responsible personalized LLM systems, and it is this interdependence that our survey aims to illuminate.